

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 5

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

Theorem

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \parallel p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}].$$

Revisiting the ELBO

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi,\theta}(\mathbf{x}_i) &= \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} \\ &\quad - \underbrace{\mathbb{I}_q[\mathbf{x}, \mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z}))}_{\text{Marginal KL}}. \end{aligned}$$

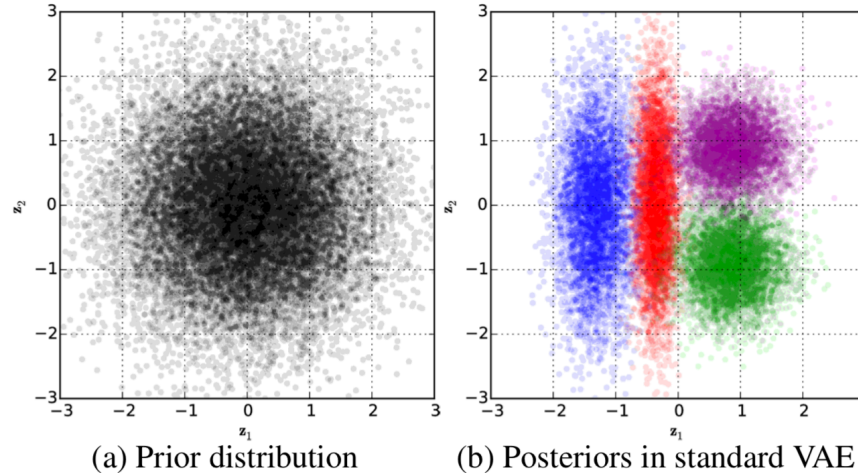
Optimal VAE Prior

$$D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) = 0 \Leftrightarrow p(\mathbf{z}) = q_{\text{agg},\phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i).$$

Thus, the optimal prior distribution $p(\mathbf{z})$ is the aggregated variational posterior $q_{\text{agg},\phi}(\mathbf{z})$.

Recap of Previous Lecture

- **Prior mismatch:** The unimodal encoder $q_\phi(\mathbf{z}|\mathbf{x})$ yields $q_{\text{agg},\phi}(\mathbf{z})$ that often does not match the Gaussian prior $p(\mathbf{z})$.
- **Blurriness from averaging:** With a fixed encoder, prior and Gaussian decoder variance, ELBO maximization gives $\mu^*(\mathbf{z}) = \mathbb{E}_{q_\phi(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$. If distinct inputs $\mathbf{x} \neq \mathbf{x}'$ map to *overlapping* latent regions, the decoder averages over unrelated data.



Recap of Previous Lecture

Assumptions

- Let $c \sim \text{Cat}(\boldsymbol{\pi})$, where

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^K \pi_k = 1.$$

- For tokenizer training, fix $p(c) = \text{Uniform}\{1, \dots, K\}$.

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x}|c) - D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) \rightarrow \max_{\phi, \theta}.$$

$$D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) = -\text{H}(q_{\phi}(c|\mathbf{x})) + \log K.$$

Quantized Representation

Define the codebook (dictionary) space $\{\mathbf{e}_k\}_{k=1}^K$ with $\mathbf{e}_k \in \mathbb{R}^L$ and K the number of codebook entries.

$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg \min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

Recap of Previous Lecture

Quantize each of the $M = WH$ encoder vectors using the **same codebook**; the code map is $\mathbf{c} = (c_1, \dots, c_M)$, with $\mathbf{z}_{q,m} = \mathbf{e}_{c_m}$.

Deterministic Variational Posterior

$$q_{\phi}(c_m = k | \mathbf{x}) = \begin{cases} 1, & \text{for } k = \arg \min_j \|\mathbf{z}_{e,m} - \mathbf{e}_j\|; \\ 0, & \text{otherwise.} \end{cases}$$

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(\mathbf{c} | \mathbf{x})} \log p_{\theta}(\mathbf{x} | \mathbf{z}_q(\mathbf{c})) - M \log K = \log p_{\theta}(\mathbf{x} | \mathbf{z}_q) - M \log K.$$

Straight-Through Gradient Estimator

$$\frac{\partial \log p(\mathbf{x} | \mathbf{z}_q, \theta)}{\partial \phi} = \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \phi} \approx \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \phi}$$

The independent uniform prior is used during tokenizer training. Then fix the tokenizer and decoder, and fit an autoregressive prior $p_{\psi}(\mathbf{c})$ to the aggregated posterior $q_{\text{agg}, \phi}(\mathbf{c})$.

Recap of Previous Lecture

Likelihood-Free Learning

- Likelihood isn't always a suitable metric for evaluating generative models.
- Sometimes, the likelihood function can't even be computed exactly.

Imagine we have two sets of samples:

- $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\theta}(\mathbf{x})$ — generated (fake) samples.

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\theta}(\mathbf{x}))$$

Assumption

The generative model $p_{\theta}(\mathbf{x})$ matches $p_{\text{data}}(\mathbf{x})$ if a discriminative model $p(y|\mathbf{x})$ can't distinguish between them — that is, if $p(y = 1|\mathbf{x}) = 0.5$ for every $\mathbf{x}$.

- **Generator:** a generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ that produces more realistic samples.
- **Discriminator:** a classifier $D_{\phi}(\mathbf{x}) \in [0, 1]$ distinguishing real from generated samples.

Recap of Previous Lecture

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log p(y = 0|\mathbf{x}) \right]$$

- **Discriminator:** A classifier $p_{\phi}(y = 1|\mathbf{x}) = D_{\phi}(\mathbf{x}) \in [0, 1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \right]$$

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

Recap of Previous Lecture

Theorem

The following minimax game

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

Reality

- Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- Generator and discriminator losses typically oscillate during GAN training.

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01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

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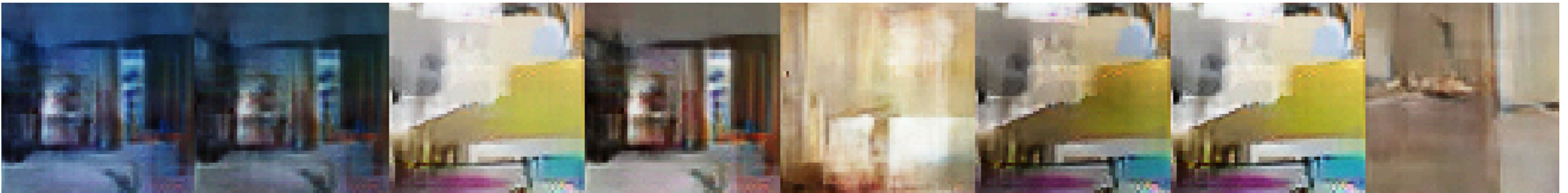
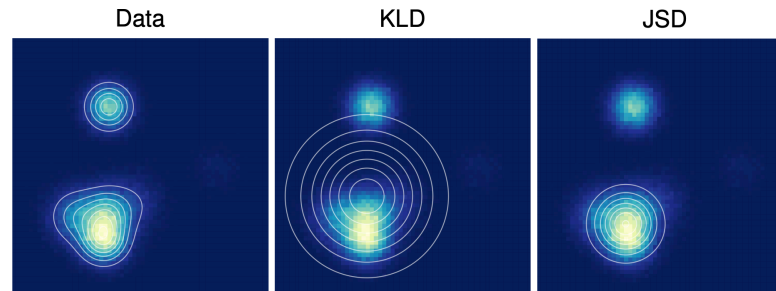
04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

Mode Collapse

Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.



Numerous methods have been proposed to tackle mode collapse: changing architectures, adding regularization terms, injecting noise.

Jensen-Shannon vs Kullback-Leibler Divergences

- $p_{\text{data}}(\mathbf{x})$ is a fixed mixture of two Gaussians.
- $p(\mathbf{x}|\mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$.

Mode Covering vs. Mode Seeking

$$D_{\text{KL}}(\pi \parallel p) = \int \pi(\mathbf{x}) \log \frac{\pi(\mathbf{x})}{p(\mathbf{x})} d\mathbf{x}, \quad D_{\text{KL}}(p \parallel \pi) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$$
$$D_{\text{JS}}(\pi \parallel p) = \frac{1}{2} \left[D_{\text{KL}}\left(\pi(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) + D_{\text{KL}}\left(p(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) \right]$$

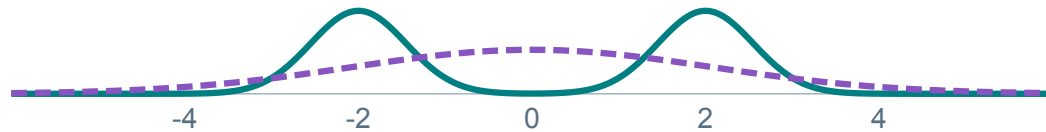
- **Forward KL** averages over data; **reverse KL** averages over model samples.
- **JSD** compares both distributions to their equally weighted mixture.

Next: fit one Gaussian with the two KL directions.

Mode Covering vs. Mode Seeking

$$p_{\text{data}}(x) = \frac{1}{2}\mathcal{N}(x \mid -2, 0.55^2) + \frac{1}{2}\mathcal{N}(x \mid 2, 0.55^2), \quad p_{\theta}(x) = \mathcal{N}(x \mid \mu, \sigma^2).$$

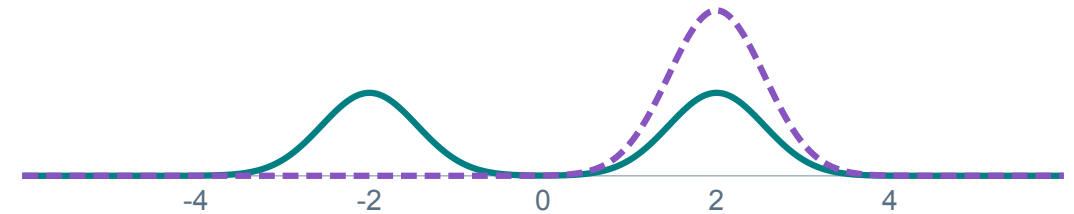
Forward KL optimum



$\mu = 0.00, \sigma = 2.07$

Forward KL 0.635 · Reverse KL 1.623

Reverse KL minimum (right mode)



$\mu = 2.00, \sigma = 0.55$

Forward KL 12.445 · Reverse KL 0.693

— Target: two modes

- - - - - Model: one Gaussian

Forward KL

Missing a target mode is expensive.

Reverse KL

Putting model mass in the low-density gap is expensive.

This difference comes from restricting the model to one Gaussian. If it could match the target exactly, both KLs would be zero.

From VQ-VAE to VQGAN

Same **encoder** → **quantization** → **decoder**; a new reconstruction objective.



Same spatial downsampling $f = 8$ and codebook size $K = 8192$.

Perceptual Loss

A **fixed pretrained network** F extracts features:

$$\mathcal{L}_{\text{perc}} = \|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|_2^2.$$

Adversarial Loss

Train discriminator D to separate real and reconstructed **image patches**.

Train the autoencoder to **fool** D .

Generative Modeling in Latent Space

First learn a compact representation; then learn to **generate its latent codes**.

Discrete: VQGAN

Quantization gives a map of code indices.

$$\mathbf{c} \in \{1, \dots, K\}^{W \times H}$$

Autoregressive prior over the code map.

$$\mathbf{c} \sim p_\psi(\mathbf{c}) \longrightarrow \mathbf{z}_q(\mathbf{c}) \longrightarrow \hat{\mathbf{x}}$$

Sample codes $\rightarrow$ lookup $\rightarrow$ decode.

Continuous: KL-Autoencoder

Perceptual + adversarial losses; weak KL.

$$\mathbf{z} \in \mathbb{R}^{W \times H \times L}$$

Diffusion prior in the learned latent space.

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \longrightarrow \mathbf{z} \sim p_\psi(\mathbf{z}) \longrightarrow \hat{\mathbf{x}}$$

Denoise in latent space $\rightarrow$ decode.

Both recipes freeze the autoencoder before learning a prior that models its **aggregated posterior**.

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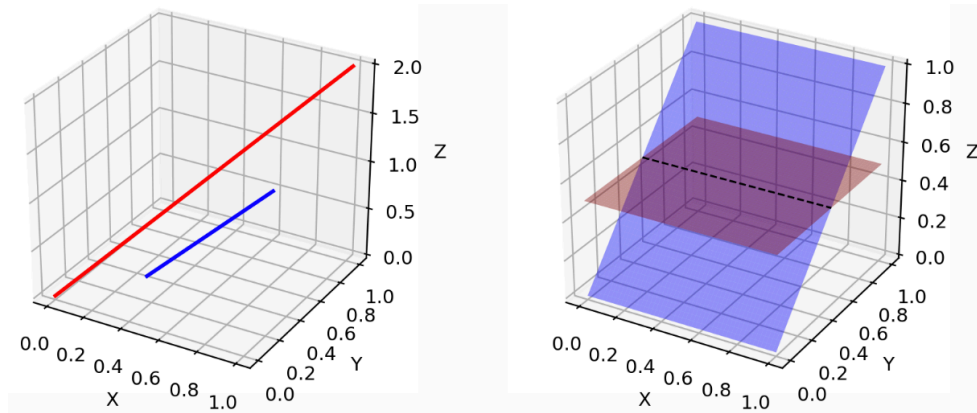
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Theoretical Results

- The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_{\theta}(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ lives on a low-dimensional manifold.
- The true data distribution $p_{\text{data}}(\mathbf{x})$ is also supported on a low-dimensional manifold.



- If $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

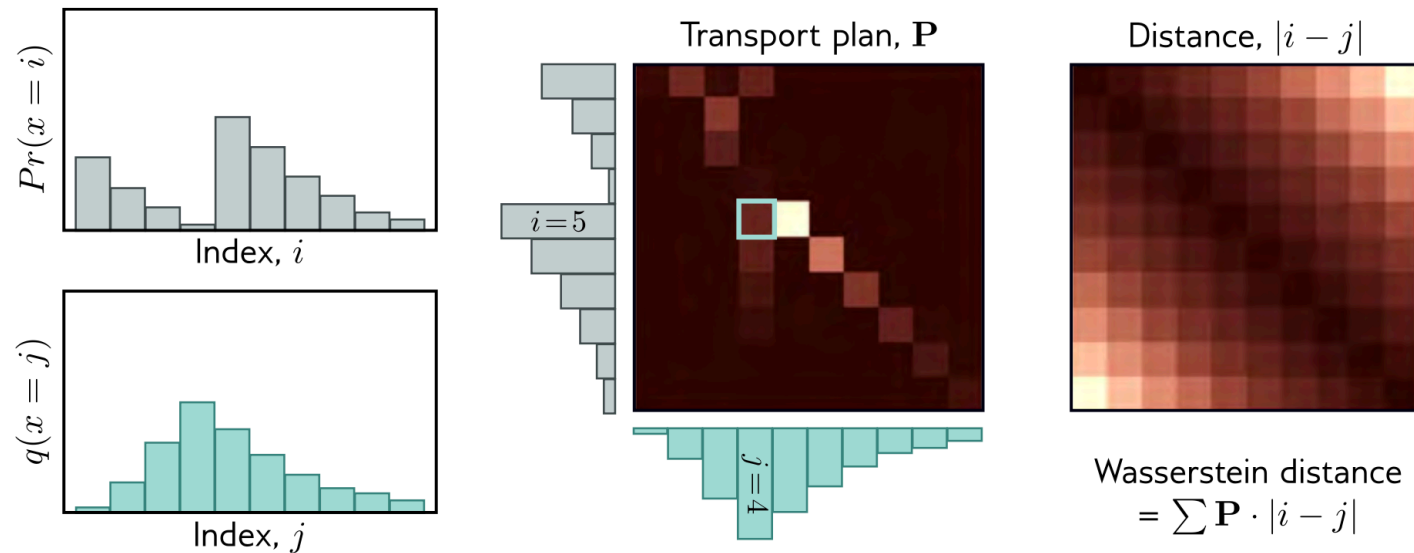
$$D_{\text{KL}}(p_{\text{data}} \parallel p_{\theta}) = D_{\text{KL}}(p_{\theta} \parallel p_{\text{data}}) = \infty, \quad D_{\text{JS}}(p_{\text{data}} \parallel p_{\theta}) = \log 2$$

Wasserstein Distance (Discrete)

Also known as the **Earth Mover's Distance**.

Optimal Transport Formulation

The minimum cost of moving and transforming a pile of “dirt” shaped like one probability distribution to match another.



Wasserstein Distance (Continuous)

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of “dirt” assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.

$$\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2); \quad \int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1).$$

- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p .
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Wasserstein Metric

$$W_s(\pi, p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

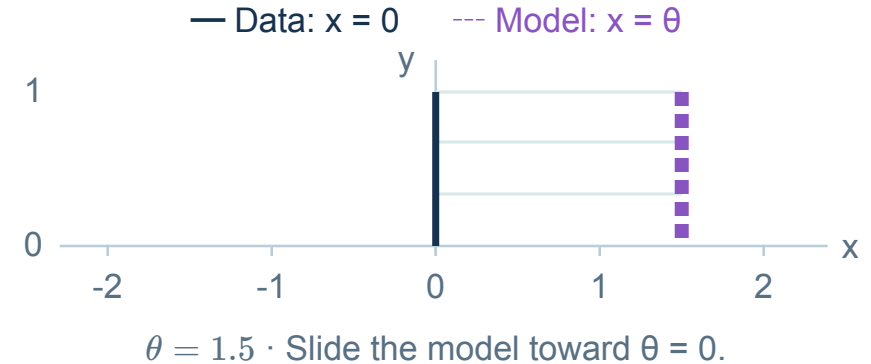
In our setting, $W(\pi \| p) = W_1(\pi, p)$, which is the transport cost formulation.

In Lecture 12, minibatch optimal transport pairs noise and data samples for flow matching.

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1]), \quad p_{\theta}(x, y) = (\theta, U[0, 1])$$



$\theta = 0$: Both distributions are identical.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

$\theta \neq 0$:

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int_{U[0,1]} 1 \log \frac{1}{0} dy = \infty = D_{\text{KL}}(p_{\theta} \| p_{\text{data}})$$

$$D_{\text{JS}}(p_{\text{data}} \| p_{\theta}) = \frac{1}{2} \left(\int_{U[0,1]} 1 \log \frac{1}{1/2} dy + \int_{U[0,1]} 1 \log \frac{1}{1/2} dy \right) = \log 2$$

$$W(p_{\text{data}} \| p_{\theta}) = |\theta|$$

Why Wasserstein Distance?

On a compact data space $\mathcal{X}$:

Dependence on Generator Parameters

For a feedforward generator G_θ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})} \|\mathbf{z}\| < \infty$, $W(p_{\text{data}} \| p_\theta)$ is continuous in θ and differentiable almost everywhere.

Convergence of Distributions

For a fixed distribution π and a sequence p_t on $\mathcal{X}$, as $t \rightarrow \infty$:

$$D_{\text{KL}}(\pi \| p_t) \rightarrow 0 \implies D_{\text{JS}}(\pi \| p_t) \rightarrow 0 \implies W(\pi \| p_t) \rightarrow 0$$

The first implication also holds for $D_{\text{KL}}(p_t \| \pi) \rightarrow 0$; the converses need not hold.

Wasserstein can capture distributions getting closer even when their supports do not overlap.

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Wasserstein GAN

Wasserstein Distance

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

The infimum over all possible $\gamma \in \Gamma(\pi, p)$ is computationally intractable.

Theorem (Kantorovich-Rubinstein Duality)

$$W(\pi \| p) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{\pi(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p(\mathbf{x})} f(\mathbf{x}) \right]$$

where $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is K -Lipschitz ($\|f\|_L \leq K$):

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq K \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}.$$

We can thus estimate $W(\pi \| p)$ using only samples and a function f .

Wasserstein GAN

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\theta}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right]$$

- We must ensure that f is K -Lipschitz continuous.
- Let $f_{\phi}(\mathbf{x})$ be a feedforward neural network parameterized by ϕ .
- If the weights ϕ are restricted to a compact set Φ , then f_{ϕ} is K -Lipschitz.
- Clamp weights within the box $\Phi = [-c, c]^d$ (e.g. $c = 0.01$) after each update.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\theta}) &= \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right] \\ &\geq \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f_{\phi}(\mathbf{x}) \right] \end{aligned}$$

Wasserstein GAN

Standard GAN Objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})))$$

WGAN Objective

$$\min_{\theta} W(p_{\text{data}} || p_{\theta}) \approx \min_{\theta} \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})) \right]$$

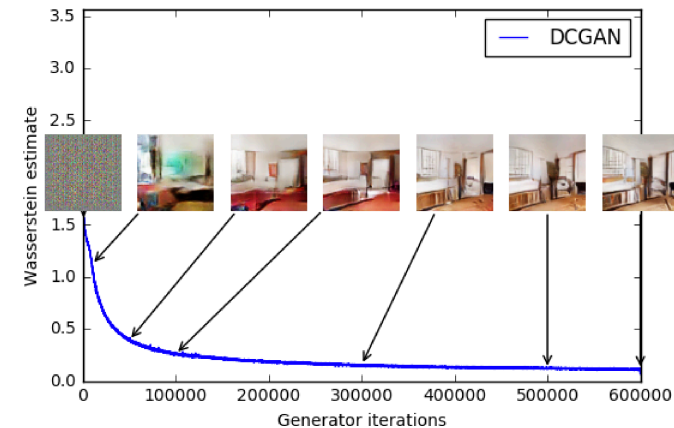
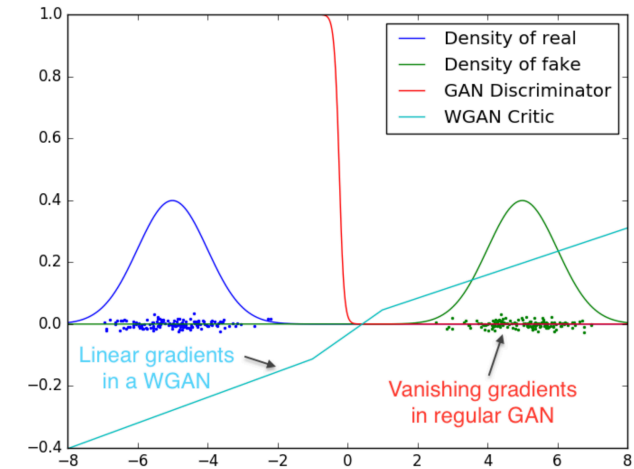
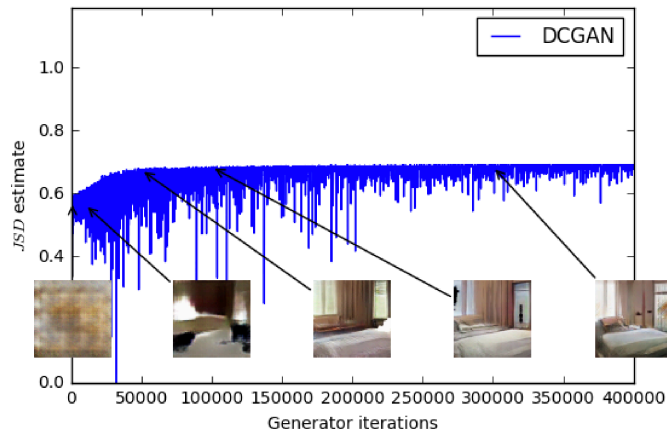
- The discriminator D is replaced by function f : in WGAN, it is known as the **critic**, which is *not* a classifier.
- “*Weight clipping is a clearly terrible way to enforce a Lipschitz constraint.*”
 - If c is large, optimizing the critic is hard; if c is small, gradients may vanish.

Gradient penalty encourages the critic’s input-gradient norm to stay close to 1 on interpolations between real and generated samples.

Spectral normalization controls layer-wise Lipschitz constants by dividing weight matrices by their largest singular values.

Wasserstein GAN

- WGAN provides nonzero gradients even if distributions' supports are disjoint.
- $D_{JS}(p_{\text{data}} || p_{\theta})$ is poorly correlated with sample quality and remains near its maximum value $\log 2 \approx 0.69$.
- $W(p_{\text{data}} || p_{\theta})$ is tightly correlated with quality.



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What Makes a Good Generator?

Likelihood alone does not tell us whether samples are useful for the target task.

Fidelity

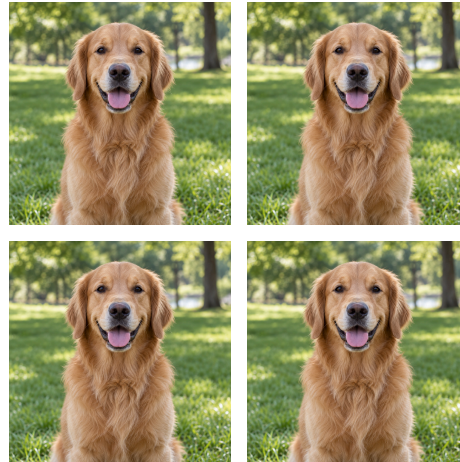
Does each sample look plausible?



Sharp, but anatomically wrong.

Coverage & Diversity

Does the set cover the data?



The same output on every run.

Condition Alignment

Does it follow the condition?



Prompt: “Three red apples.”
Result: only two.

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Wasserstein Metric

$$W_s(\pi||p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

Wasserstein GAN (Optimal Transport)

$$\begin{aligned} W(\pi||p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

Theorem

If $\pi(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\pi, \boldsymbol{\Sigma}_\pi)$, $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, then

$$W_2^2(\pi||p) = \|\boldsymbol{\mu}_\pi - \boldsymbol{\mu}_p\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_\pi + \boldsymbol{\Sigma}_p - 2 \left(\boldsymbol{\Sigma}_\pi^{1/2} \boldsymbol{\Sigma}_p \boldsymbol{\Sigma}_\pi^{1/2} \right)^{1/2} \right]$$

Frechet Inception Distance

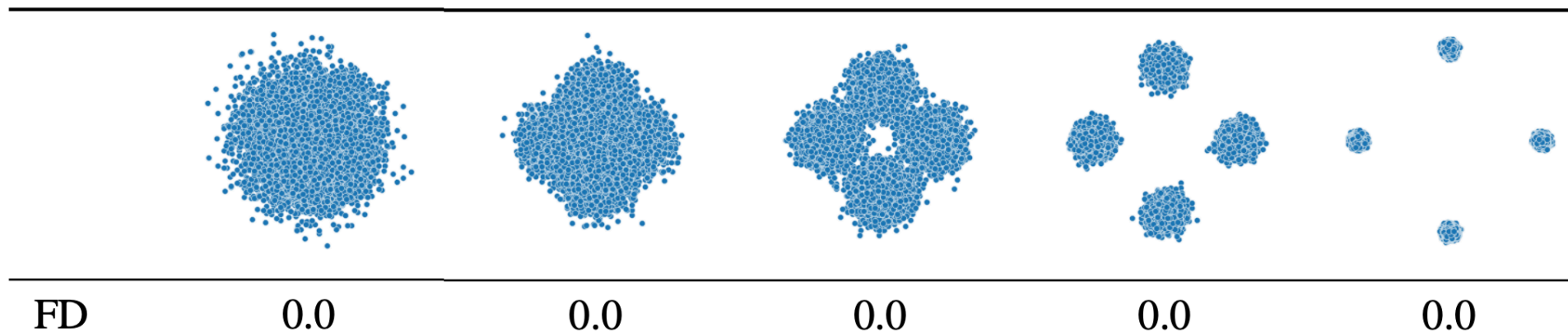
$$\text{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = W_2^2(p_{\text{data}}||p_{\boldsymbol{\theta}})$$

Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\theta}\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\theta} - 2 \left(\boldsymbol{\Sigma}_{\text{data}}^{1/2} \boldsymbol{\Sigma}_{\theta} \boldsymbol{\Sigma}_{\text{data}}^{1/2} \right)^{1/2} \right]$$

- FID uses Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$.
- $\boldsymbol{\mu}_{\text{data}}$, $\boldsymbol{\Sigma}_{\text{data}}$ and $\boldsymbol{\mu}_{\theta}$, $\boldsymbol{\Sigma}_{\theta}$ are statistics of latent representations for samples from $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$.
- **FD-DINOv2** uses the same formula with DINOv2 features; the Gaussian approximation remains.

$\text{FID}(p(\mathbf{x}), \mathcal{N}(0, \mathbf{I}))$



Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\mu_{\text{data}} - \mu_{\theta}\|^2 + \text{tr} \left[\Sigma_{\text{data}} + \Sigma_{\theta} - 2 \left(\Sigma_{\text{data}}^{1/2} \Sigma_{\theta} \Sigma_{\text{data}}^{1/2} \right)^{1/2} \right]$$

Drawbacks

- Depends on the pretrained classification network.
- Uses the normality assumption.
- May not correlate with human evaluation.

Finite-sample FID is biased, and its value depends on the sample size.

Model	Model-A	Model-B
FID	21.40	18.42
FID _∞	20.16	17.19
Human rater preference	92.5%	6.9%

Maximum Mean Discrepancy (MMD)

The earlier figure shows that matching means and covariances can miss changes in shape. For embedding distributions P, Q , compare means of richer nonlinear features ψ :

$$\boldsymbol{\mu}_P = \mathbb{E}_{\mathbf{z} \sim P} \psi(\mathbf{z}), \quad \boldsymbol{\mu}_Q = \mathbb{E}_{\mathbf{w} \sim Q} \psi(\mathbf{w}).$$

Kernel Trick

$$k(\mathbf{a}, \mathbf{u}) = \langle \psi(\mathbf{a}), \psi(\mathbf{u}) \rangle.$$

Compute inner products directly, without explicitly constructing ψ .

Distance Between Mean Features

Let $\mathbf{z}, \mathbf{z}' \sim P$ and $\mathbf{w}, \mathbf{w}' \sim Q$ be mutually independent.

$$\begin{aligned} \text{MMD}_k^2(P, Q) &= \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_Q\|^2 \\ &= \langle \boldsymbol{\mu}_P, \boldsymbol{\mu}_P \rangle + \langle \boldsymbol{\mu}_Q, \boldsymbol{\mu}_Q \rangle - 2\langle \boldsymbol{\mu}_P, \boldsymbol{\mu}_Q \rangle \\ &= \underbrace{\mathbb{E}k(\mathbf{z}, \mathbf{z}')}_{\text{within } P} + \underbrace{\mathbb{E}k(\mathbf{w}, \mathbf{w}')}_{\text{within } Q} - 2 \underbrace{\mathbb{E}k(\mathbf{z}, \mathbf{w})}_{\text{between } P, Q}. \end{aligned}$$

CMMD: Comparing CLIP Image Embeddings

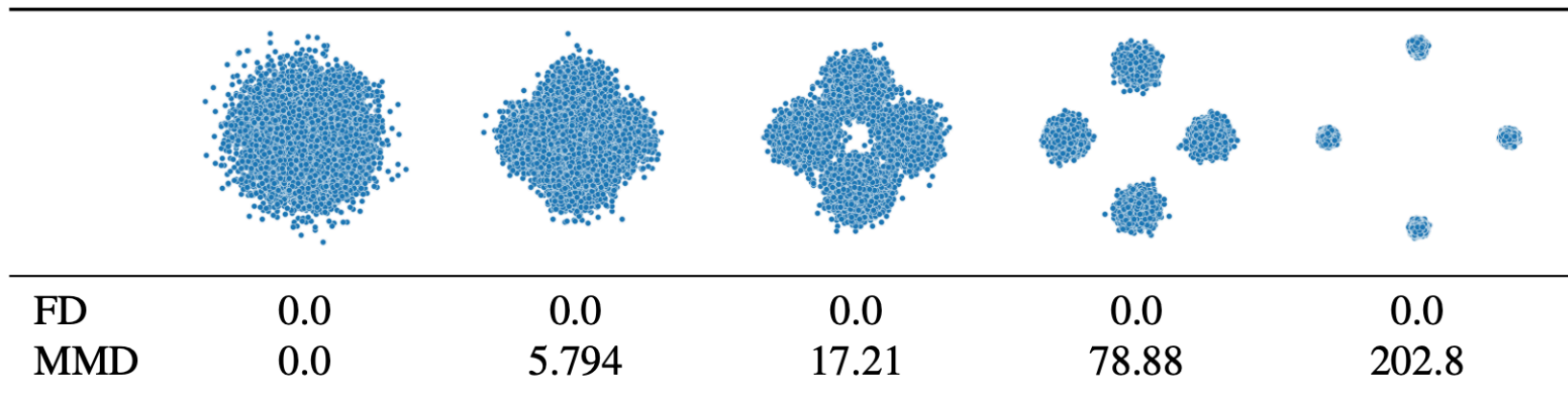
CMMD = squared MMD on **CLIP image embeddings**.

$$\text{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

$$k(\mathbf{z}, \mathbf{w}) = \exp\left(-\frac{\|\mathbf{z} - \mathbf{w}\|^2}{2\sigma^2}\right), \quad \sigma > 0.$$

One Gaussian kernel already corresponds to infinitely many nonlinear features in ψ .

$$\text{MMD}_k(P, Q) = 0 \iff P = Q.$$



Synthetic example: same means and covariances, different shapes.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

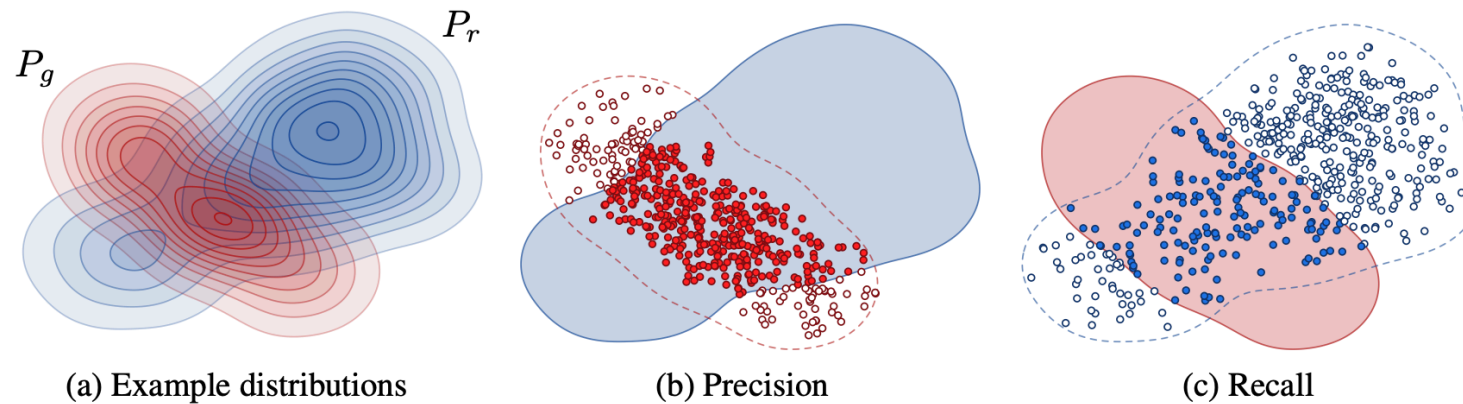
Frechet Inception Distance (FID) · **Precision-Recall** · CLIP Score · Human Eval

05 Langevin Dynamics

Precision-Recall

Desirable Properties for Samples

- **Sharpness:** generated samples should possess high visual quality.
- **Diversity:** their variation should match that in the training data.



- **Precision** denotes the fraction of generated images that look realistic.
- **Recall** measures how well the generator covers the training data manifold.

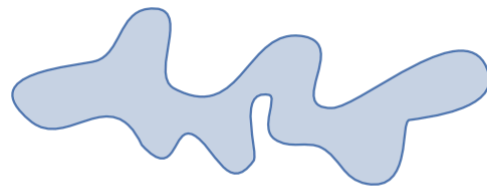
Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ — generated samples.

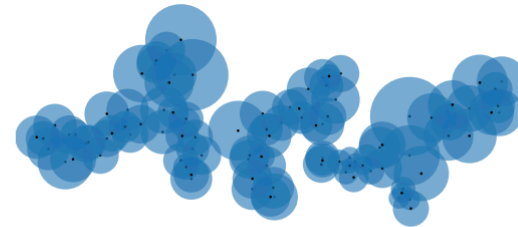
Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\theta}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\theta}).$$



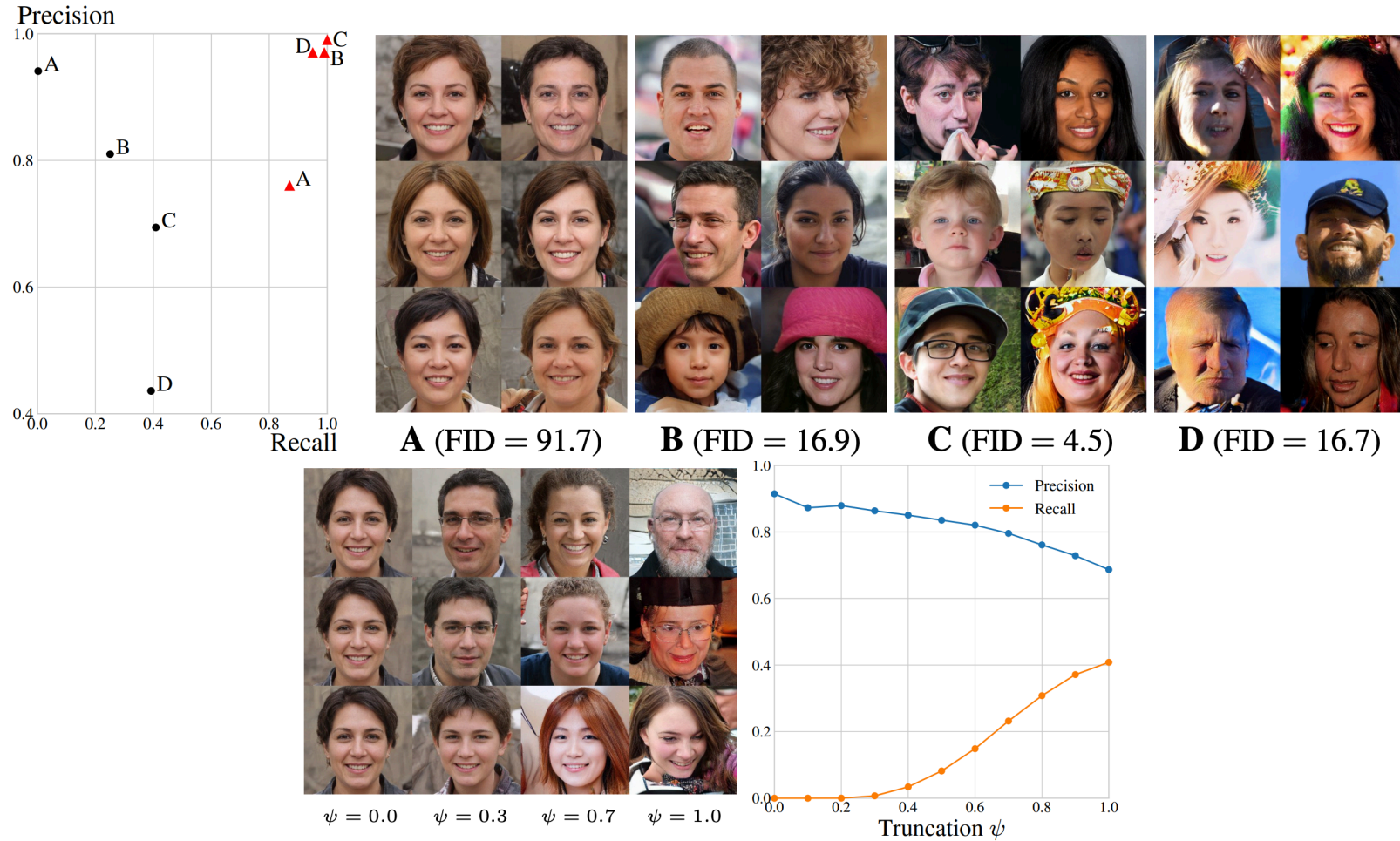
(a) True manifold



(b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

Precision-Recall



Here, stronger truncation (smaller ψ) favors precision over recall.

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CLIP Similarity and Human Preferences

Score an image $\mathbf{x}$ for a text prompt t .

CLIP: Learn from Image–Caption Pairs

Map images and text into a shared embedding space:

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)).$$

PickScore: Learn from Human Choices

Fine-tune CLIP on pairs: for the same prompt, a human prefers $\mathbf{x}^+$ to $\mathbf{x}^-$.

$$\mathbb{P}_{\theta}(\mathbf{x}^+ \succ \mathbf{x}^- \mid t) = \frac{\exp r_{\theta}(\mathbf{x}^+, t)}{\exp r_{\theta}(\mathbf{x}^+, t) + \exp r_{\theta}(\mathbf{x}^-, t)}.$$

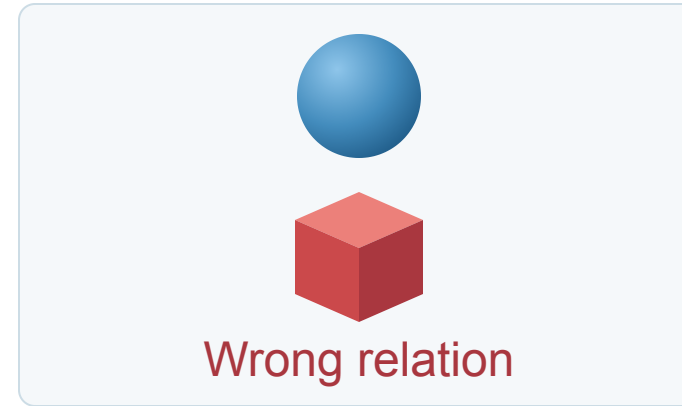
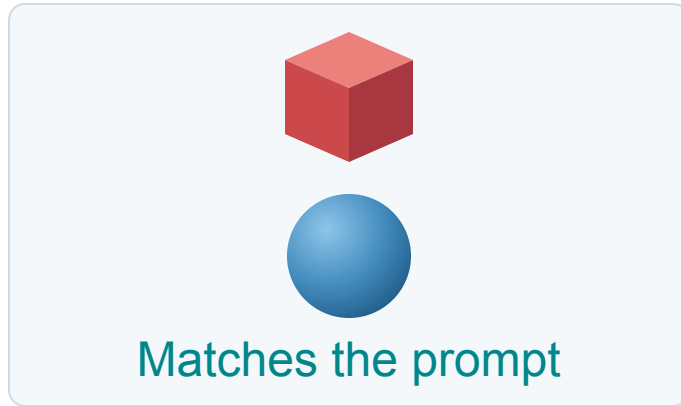
Train the score r_{θ} to assign higher probability to the human choice.

Other preference models: [ImageReward](#), [HPSv2](#).

A learned preference score can guide sample selection and model training — but optimizing the score can exploit its flaws.

Does the Image Follow the Prompt?

Prompt t : “A red cube above a blue sphere.”



Same objects and colors; opposite spatial relation. CLIP similarity can miss this.

VQAScore: Ask a Visual Question

$q(t)$: “Does this image show a red cube above a blue sphere?”

$$s_{\text{VQA}}(\mathbf{x}, t) = p_{\text{VQA}}(\text{Yes} \mid \mathbf{x}, q(t)).$$

GenEval / DPG-Bench: test objects, counts, attributes, and relations across prompts.

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Human Evaluation

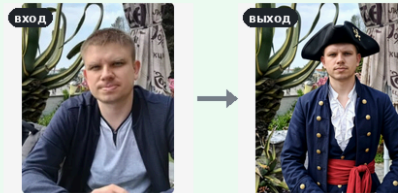
- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It's important to assess various properties.

РЕЛЕВАНТНОСТЬ

«Выполнена ли инструкция?»

Сделал ли редактор именно то, что просили

✓ ХОРОШО



Запрос: «сделать пиратом». Образ пирата создан — то, о чём просили, выполнено.

✗ ПЛОХО



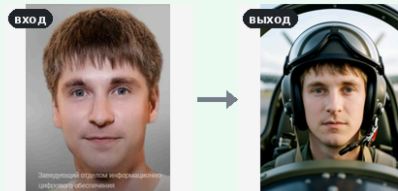
Запрос: эффект вращения на карусели. Картинка почти не изменилась — инструкция не выполнена.

СОХРАНЕНИЕ

«Сохранилось ли то, что менять не просили?»

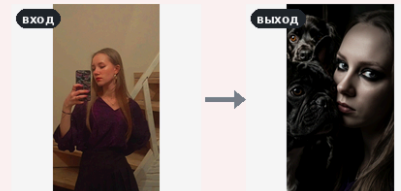
Фон, лицо и композиция остаются на месте

✓ ХОРОШО



Запрос: образ лётчика, «не меняя черты лица». Под шлемом — то же самое лицо.

✗ ПЛОХО



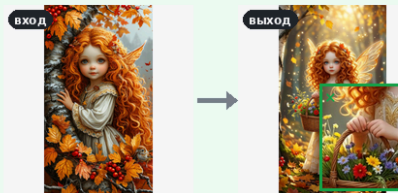
Запрос: «не меняй внешность, один в один», лишь приблизить лицо к собаке. Но на выходе — совершенно другой человек.

ДЕФЕКТЫ

«Не появилось ли артефактов?»

Лишние пальцы, сломанная геометрия и т. п.

✓ ХОРОШО



Кисти рук и предметы без искажений — артефактов нет.

✗ ПЛОХО



Кисть руки слилась в бесформенный ком (увеличено) — явный артефакт.

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Energy-Based Models

Unnormalized Density

$$p_{\theta}(\mathbf{x}) = \frac{\hat{p}_{\theta}(\mathbf{x})}{Z_{\theta}}, \quad \text{where } Z_{\theta} = \int \hat{p}_{\theta}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\theta}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\theta}(\mathbf{x}) = \exp(-f_{\theta}(\mathbf{x}))$, we eliminate the non-negativity constraint.

Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\theta} = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x})$$

- Suppose we already have this density (normalized or not) $p_{\theta}(\mathbf{x})$.
- How can we sample from the model?

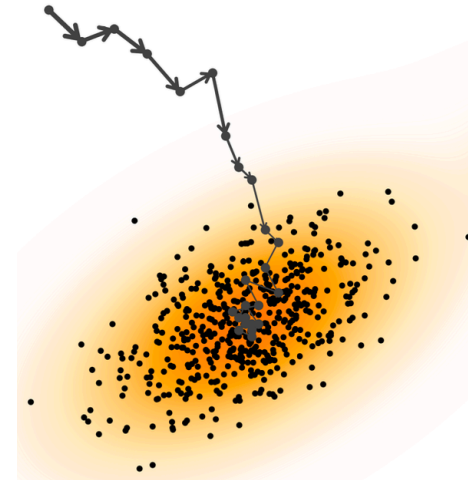
Langevin Dynamics

Theorem (Informal)

Let $\mathbf{x}_0$ be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow $p_\theta(\mathbf{x})$ (for sufficiently small η and large l):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_\theta(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

- What if $\boldsymbol{\epsilon}_l = \mathbf{0}$?
- The density $p_\theta(\mathbf{x})$ is the **stationary** distribution of the Markov chain.
- The gradient is taken with respect to $\mathbf{x}$, not $\boldsymbol{\theta}$.
- $\nabla_{\mathbf{x}} \log p_\theta(\mathbf{x})$ defines a vector field.



Langevin Dynamics: Why Add Noise?

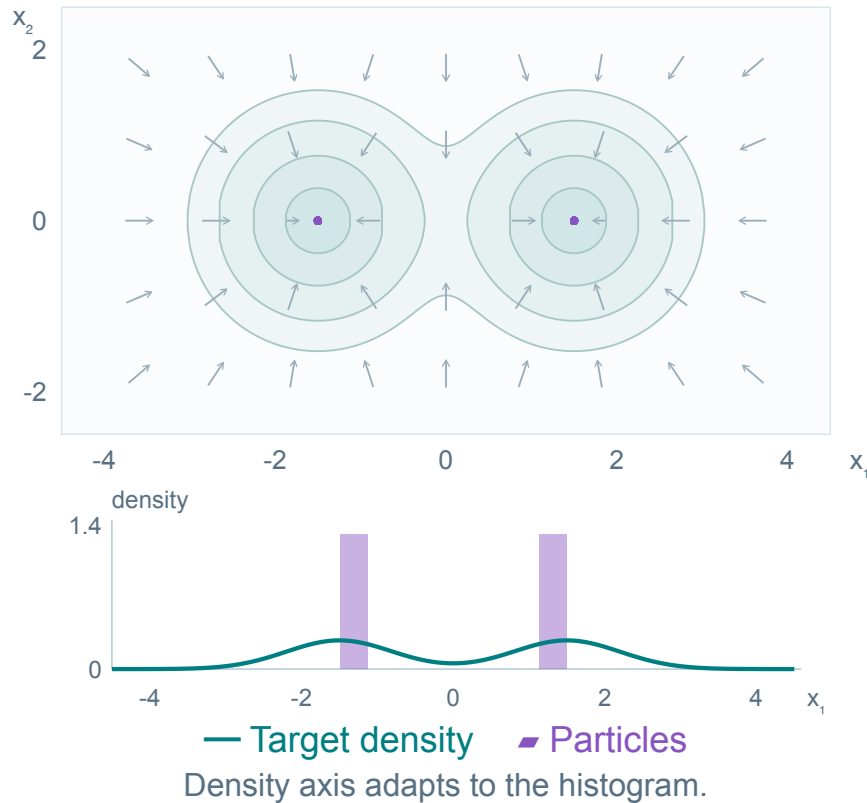
$$p_{\theta}(\mathbf{x}) = \frac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{-}, 0.7^2\mathbf{I}) + \frac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{+}, 0.7^2\mathbf{I}), \quad \boldsymbol{\mu}_{\pm} = (\pm 1.5, 0), \quad \eta = 0.04.$$

Noise off: approach the modes

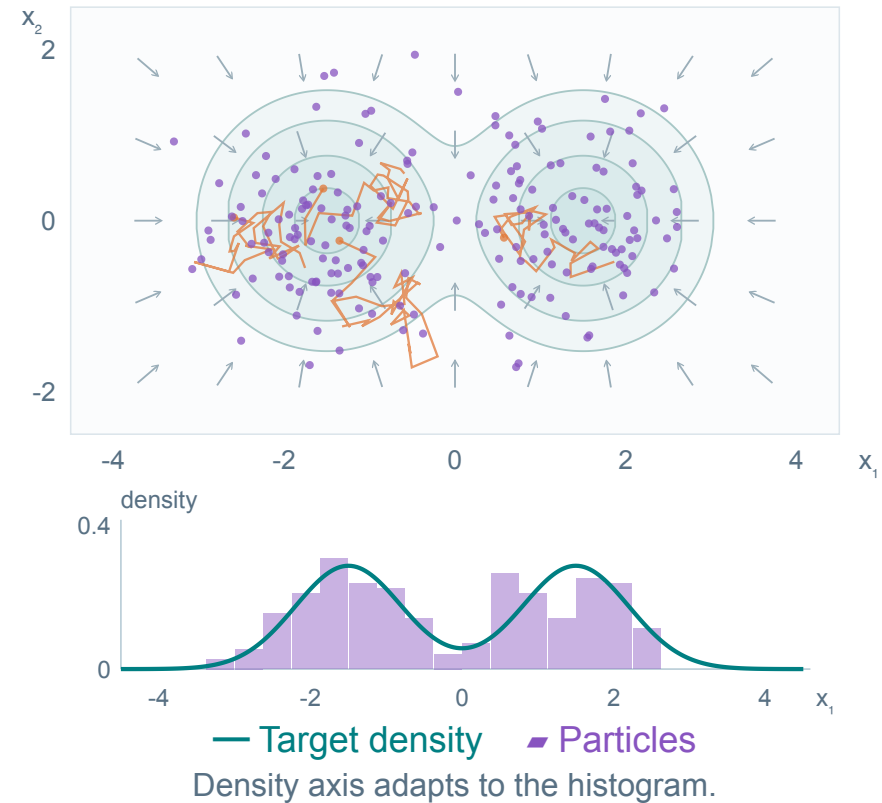
Step 300

Noise on: explore the density

Step 300



Finite step size and finite simulation time give approximate sampling.



Summary

- GANs can suffer from mode collapse; KL and Jensen-Shannon measure mismatch differently.
- Perceptual and adversarial losses train autoencoders; learned priors model their latents.
- WGAN uses Kantorovich-Rubinstein duality to estimate a distance informative even for disjoint supports; weight clipping constrains the critic.
- FID compares Gaussian approximations; CMMD compares distributions of CLIP features using a kernel.
- Precision-recall measures the balance between sample quality and diversity.
- Image–text similarity, human preference, and compositional correctness require different evaluation signals.
- Langevin dynamics uses log-density gradients and Gaussian noise to sample from an unnormalized model.